SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Morphological Inflection Generation

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Morphological Inflection Generation

Titel der Abschlussarbeit

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Submission Date:	Submission date

I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, Submission date | Murilo Escher Pagotto Ronchi

Acknowledgments

Abstract

Contents

1 Introduction

Citation test (Lamport, 1994).

Acronyms must be added in `main.tex` and are referenced using macros. The first occurrence is automatically replaced with the long version of the acronym, while all subsequent usages use the abbreviation.

E.g. `\ac{TUM}, \ac{TUM}` ⇒ Technical University of Munich (TUM), TUM

For more details, see the documentation of the `acronym` package[1].

See Table 1.1, Figure 1.1, Figure 1.2, Figure 1.3.

Table 1.1: An example for a simple table.

A	B	C	D
1	2	1	2
2	3	2	3

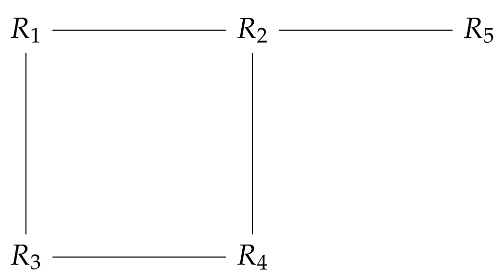

Figure 1.1: An example for a simple drawing.

[1] `https://ctan.org/pkg/acronym`

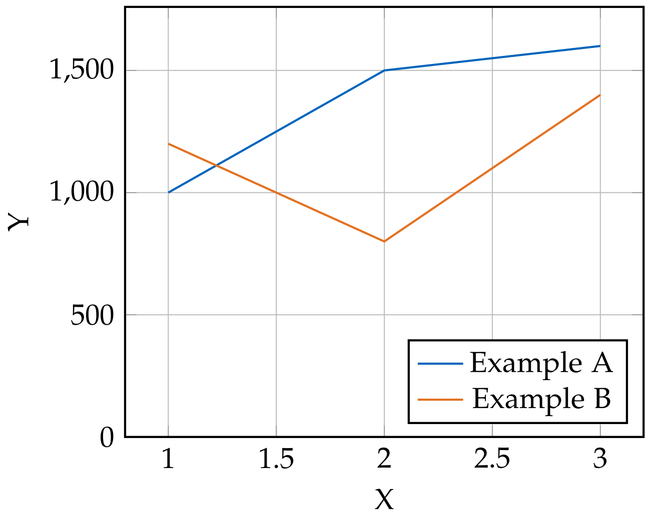

Figure 1.2: An example for a simple plot.

```
SELECT * FROM tbl WHERE tbl.str = "str"
```

Figure 1.3: An example for a source code listing.

2 Background

Different models have been used in this thesis to tackle two different, but related tasks. This chapter presents both tasks, the theoretical background and functionality behind the models used, as well as explanations of linguistic concepts pertaining to the tasks.

2.1 Tasks

This thesis tackles two related problems, which can be seen as the opposite linguistic operation of one another. Both challenges, presented in the following two subsections, relate to the operation of applying a set of linguistic transformations on top of a base form of a word to achieve an inflected form.

A lemma, commonly known as root, represents the base form of a word (BETTER DEFINITION). This base form is subject to different transformations, comprised of different categories of change. These can be, amongst other, related to mood, aspect, tense, case, number and person. To represent these transformations, a list of morphological features is used, in this work commonly referred to as morphological tags. This list of features can then be applied on some lemma, resulting in the inflected form.

This thesis addresses two closely related problems that can be viewed as inverse linguistic operations. Both tasks, introduced in the following subsections, involve applying a set of linguistic transformations to a base form of a word in order to produce an inflected form.

A lemma, commonly referred to as the root or canonical form, denotes the base representation of a word. This base form can undergo various transformations belonging to different grammatical categories, including mood, aspect, tense, case, number, and person. These transformations are represented as a collection of morphological features, which in this work are referred to as *morphological tags*. Applying a specific set of such features to a lemma yields the corresponding inflected form.

2.1.1 Morphological Inflection

Morphological inflection is the direct form of the operation, that is, the application of the morphological features onto the lemma. This task has been studied by many different works.

An example of this task can be seen in MAKE FIGURE

This topic is not only relevant to pure linguistic research, but also presents tangible real-world benefits in computation applications. Mostly due to the fact that, starting from a lemma, given the necessary understanding of the context and the corresponding morphological features that would need to be applied, it is possible to generate different inflected forms of various base forms. This represents a clear advantage in the reduced necessity of saving different word variations, lowering the memory requirements for a dictionary of words present in an application.

2.1.2 Morphological Analysis

Morphological analysis can be seen as the reverse operation to morphological inflection. It consists in taking an inflected form of some word and separating it into lemma and morphological features.

An example of morphological analysis can be seen in FIGURE CITE

This task PRESENT USE-CASES FOR THIS; WHICH MIGHT NOT BE CLEAR; GOTTA READ

2.2 Inflection Baselines

2.2.1 Non-Neural Baseline

The first model evaluated was the non-neural baseline provided in the Sigmorphon 2023 shared task Goldman et al., 2023 repository. This is a statistical model based on the work first presented in Cotterell et al., 2017, which uses a set of edit rules extracted from the training data to perform morphological inflection generation.

This rule-based system learns a set of prefix and suffix transformation rules from the training data. For each morphological tag (MSD), the model extracts the possible prefix and suffix transformation rules that map the lemma to the inflected form. During inference, given a lemma and a target MSD, the model applies the most frequent rules associated with the specific transformation to generate the inflected form.

For example, given the Portuguese verb *andar* (to walk) and the target MSD `V;PRS;3;SG` (verb, present tense, third person singular), the model might learn the suffix transformation rule `AR -> A` from the training data. Applying this rule to the lemma *amar* would yield the inflected form *ama*.

How the model works, is that it first extracts all possible prefix and suffix transformation rules from the training data. For each lemma-inflected form pair, it identifies the longest common prefix and suffix, and derives the transformation rules by comparing

Verb	Target MSD	Inflected Form	Transformation Rule
andar	`V;PRS;3;SG`	*anda*	`AR -> A`
amar	`V;PRS;3;SG`	*ama*	`AR -> A`

Table 2.1: Example of transformation rule learned by the non-neural baseline

the remaining parts of the lemma and inflected form. The model keeps track of the frequency of each rule for each MSD.

2.2.2 Neural Baseline

Placeholder text.

2.3 ByT5

Placeholder text.

2.4 Large Language Models

Placeholder text.

3 Datasets and Languages

3.1 Datasets

To address the tasks of morphological inflection and morphological analysis and to train the proposed models, labeled data is required. Each instance consists of a lemma, a set of morphological tags describing the target grammatical features, and the corresponding inflected form. For this purpose, this work relies on two widely used resources: UniMorph 4.0 (Batsuren et al., 2022) and Universal Dependencies (UD) (Nivre et al., 2020).

3.1.1 UniMorph 4.0

UniMorph 4.0 is a multilingual project dedicated to providing large-scale morphological resources in the form of inflection tables and structured morphological annotations across a wide range of languages. At the time of writing, the dataset covers up to 182 languages, including both high-resource and low-resource as well as endangered languages.

For each language, UniMorph provides a dataset consisting of individual instances that pair a lemma with a bundle of morphological features and the corresponding inflected form. Concretely, the data is distributed as tab-separated lines, where each row represents a single lemma, tags, form triple. Multiple rows typically share the same lemma and correspond to different morphological feature combinations, effectively encoding the full inflectional paradigm in a flat, line-based format, as shown in Table 3.1.

Lemma	Tags	Inflected form
walk	V;PST;3;SG	walked

Table 3.1: Example instance of the tab-separated format used by UniMorph.

In this example, the morphological tags can be interpreted as shown in Table 3.2:

The UniMorph data served as the main resource throughout the experimental pipeline. Following the original SIGMORPHON shared task setup, it formed the

Tag	Meaning
V	Verb
PST	Past tense
3	Third person
SG	Singular

Table 3.2: Example UniMorph morphological tags and their meanings.

basis for the reimplementation of the shared task baselines.

Furthermore, the dataset was used to fine-tune the ByT5 model for morphological inflection and, by inverting the source–target mapping, also for morphological analysis. It additionally constituted the core training data for the bidirectional model, while certain experimental settings were further supplemented with data derived from Universal Dependencies.

3.1.2 Universal Dependencies

The UD resource provides consistent morphosyntactic annotation across languages, offering dependency treebanks with standardized part-of-speech tags, morphological features, and syntactic relations (Marneffe et al., 2021). At the time of writing, UD covers over 150 languages through more than 200 treebanks, spanning both high-resource and low-resource settings.

The data is distributed in the CoNLL-U format, where each sentence is represented as a sequence of token-level annotations. In addition to syntactic dependency information, each token is associated with a lemma and a set of morphological features encoded as attribute–value pairs (e.g., `Tense=Past|Number=Sing|Person=3`). Compared to UniMorph's tab-separated triple format, UD therefore provides richer contextual and sentence-level information alongside morphological annotations. An example taken from the Portuguese PetroGold treebank is shown in Figure 3.1.

In this work, UD data serves as a source of contextual information for fine-tuning the bidirectional ByT5 model and for conducting experiments with large language models.

Since UD often provides multiple treebanks for a given language, it was necessary to select a single resource for each case. To guide this choice, we relied on the ranking provided by the UD project, which evaluates treebanks based on their size, annotation quality, and linguistic diversity. Whenever multiple treebanks were available for a language, the highest-ranked treebank according to this score was selected.

```
# text = Estas podem representar falhas profundas.
# sent_id = 247-20140910-MONOGRAFIA_0-42
1 Estas este PRON _ Gender=Fem|Number=Plur|PronType=Dem 2 nsubj _ _
2 podem poder VERB _ Mood=Ind|Number=Plur|Person=3|Tense=Pres|VerbForm=Fin
0 root _ _
3 representar representar VERB _ VerbForm=Inf 2 xcomp _ _
4 falhas falha NOUN _ Gender=Fem|Number=Plur 3 obj _ _
5 profundas profundo ADJ _ Gender=Fem|Number=Plur 4 amod _ SpaceAfter=No
6 . . PUNCT _ _ 2 punct _ _
```

Figure 3.1: Example sentence in CoNLL-U format from the Portuguese UD PetroGold treebank.

3.1.3 Data Preparation

Since the two data sources use different annotation schemes and file formats, several preprocessing and conversion steps were required to obtain a unified representation suitable for all experiments.

For UniMorph, we followed the official SIGMORPHON shared task setup whenever available, using the predefined training, development, and test splits provided for the corresponding languages. These partitions consist of 10,000 training instances, 1,000 development instances, and 1,000 test instances. For languages not covered by the shared task, the data was obtained directly from the original UniMorph repository and split into training, development, and test sets using the same size configuration to maintain consistency across languages.

For most UD treebanks, predefined training, development, and test splits are provided. However, for Amharic only a test set is available. In this case, the data was randomly partitioned into training, development, and test sets according to the same proportions as the other languages, as no official split specification is provided.

As the project builds upon a reimplementation of the SIGMORPHON baselines, the final data format closely follows the UniMorph specification. Each instance therefore consists of a lemma, a set of morphological tags, and the corresponding inflected form, separated by tab characters. For data originating from 3.1, an additional fourth column containing the sentence context is included when contextual information is required.

To incorporate UD data into this format, the morphological feature annotations provided in the CoNLL-U files were first mapped to their UniMorph equivalents. For this purpose, we used the official UniMorph conversion tool, which translates UD feature sets into the UniMorph schema using language-specific translators (McCarthy et al., 2018). Since not all evaluated languages were supported by the provided translators, the conversion scripts were extended following the structure of the existing templates to additionally support Georgian and Amharic.

Figure 3.2 illustrates the output of the conversion tool when applied to the sentence shown in Figure 3.1. The UniMorph-compatible tags replace the original UD feature representations while preserving the original tokenization and syntactic structure.

```
# text = Estas podem representar falhas profundas.
# sent_id = 247-20140910-MONOGRAFIA_0-42
1 Estas este PRON _ PRO;PL;FEM 2 nsubj _ _
2 podem poder VERB _ PL;PRS;V;IND;3 0 root _ _
3 representar representar VERB _ V;NFIN 2 xcomp _ _
4 falhas falha NOUN _ PL;FEM;N 3 obj _ _
5 profundas profundo ADJ _ PL;FEM;ADJ 4 amod _ SpaceAfter=No
6 . . PUNCT _ _ 2 punct _ _
```

Figure 3.2: Example output from official conversion tool.

Since the UniMorph data predominantly contains verbs, UD data was filtered accordingly to include only verbal tokens. Concretely, for each sentence, the first verb was selected and used to construct a single instance.

After conversion, the lemma, the inflected form, the mapped UniMorph tags, and the surrounding sentence context were extracted to construct the final tab-separated instances used in the downstream experiments. The resulting unified format is illustrated in Table 3.3.

Lemma	Tags	Inflected form	Context
poder	V;IND;PRS;PL;3	podem	Estas podem representar falhas profundas.

Table 3.3: Example instance of the unified tab-separated format used in this work.

All preprocessing, conversion, and extraction scripts are included in the accompanying code repository to ensure full reproducibility.

3.2 Languages

The following tables summarize the languages considered in the experiments. Table 3.4 presents general typological and linguistic information, together with the UniMorph data source for each language, indicating whether the data was obtained from the SIGMORPHON shared task or sampled directly from the original UniMorph repository. For all languages, UniMorph uses fixed training, development, and test splits of 10,000, 1,000, and 1,000 instances, respectively. , . Additionally, the average number of possible inflected forms per lemma in the entire UniMorph dataset is reported as a coarse estimate of morphological complexity.

Table 3.5 provides statistics for the Universal Dependencies resources. Since UD treebanks are not available for Amuzgo and Lower Sorbian, these languages are excluded from the contextual and large language model experiments. For the remaining languages, the selected treebank and the sizes of the training, development, and test splits are listed.

The following tables summarize the languages considered in the experiments. Table 3.4 presents general typological and linguistic information, together with the UniMorph data source for each language, indicating whether the data was obtained from the SIGMORPHON shared task or sampled directly from the original UniMorph repository. For all languages, UniMorph uses fixed training, development, and test splits of 10,000, 1,000, and 1,000 instances, respectively. In addition, the total number of UniMorph instances ("UM size") is reported to reflect the overall amount of available training data. The average number of inflected forms per lemma is included as a coarse estimate of morphological richness. Finally, the linguistic diversity score proposed by (Joshi et al., 2020) is provided as an external measure of language resource availability and typological difficulty.

Table 3.5 provides statistics for the Universal Dependencies resources. Since UD treebanks are not available for Amuzgo and Lower Sorbian, these languages are excluded from the contextual and large language model experiments. For the remaining languages, the selected treebank and the sizes of the training, development, and test splits are listed.

Language	Family	Type	Ling. score	UM source	UM size	Avg. forms
Amharic	Semitic	Templatic	2	SIG	46224	–
Amuzgo	Oto-Manguean	Polysynthetic	0	Sampled	12204	–
Ancient Greek	Hellenic	Fusional	3	SIG	41593	–
Lower Sorbian	Slavic	Fusional	1	Sampled	20121	–
English	Germanic	Analytic	5	SIG	115523	–
Italian	Romance	Fusional	4	SIG	509574	–
Georgian	Kartvelian	Agglutinative	3	SIG	74412	–
Portuguese	Romance	Fusional	4	Sampled	303996	–

Table 3.4: Typological and UniMorph statistics for the selected languages.

Language	Treebank	Train	Dev	Test
Amharic	ATT	811	102	99
Ancient Greek	PTNK	692	424	404
English	GUM	7692	1220	1171
Italian	ISDT	10728	475	404
Georgian	GNC	798	120	741
Portuguese	PetroGold	5871	568	894

Table 3.5: Universal Dependencies treebanks and dataset sizes for the languages with available UD resources.

4 Methodology

4.1 Morphological Inflection

Placeholder text.

4.1.1 Non-Neural Baseline

Placeholder text.

4.1.2 Neural Baseline

Placeholder text.

4.1.3 ByT5

Placeholder text.

4.1.4 Bidirectional Model

Placeholder text.

4.1.5 Evaluation

Placeholder text.

4.2 Morphological Analysis

Placeholder text.

4.2.1 ByT5

Placeholder text.

4.2.2 Bidirectional Model

Placeholder text.

4.2.3 Bidirectional Model with Context

Placeholder text.

4.2.4 Large Language Models

Placeholder text.

4.2.5 Evaluation

Placeholder text.

5 Results

5.1 Morphological Inflection

Placeholder text.

5.2 Morphological Analysis

Placeholder text.

6 Conclusion

Placeholder text.

Abbreviations

TUM Technical University of Munich

UD Universal Dependencies

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